

BUSINESS ANALYSIS REPORT

Task 3: Advanced SQL Analysis, KPI Modeling & Variance Dashboard

1. Executive Summary

This report presents a comprehensive business intelligence analysis of the Sample Superstore dataset, covering the period from 2014 to 2017. The analysis was conducted using advanced SQL queries, Excel variance calculations, and an interactive Tableau dashboard. Key findings reveal strong overall revenue growth, regional performance disparities, and category-level profitability gaps that require strategic attention.

Key Highlights:

- Total Revenue: \$2.3M across all regions and categories
- Total Profit: \$286K with an overall profit margin of approximately 12.4%
- Top Performing Region: West (\$725K in sales)
- Underperforming Region: Central (\$500K, lowest profit margin)
- Highest Profit Category: Technology (\$145K profit)
- Top Customer: Sean Miller (\$25,043 in total purchases)

2. Monthly Sales Trend Analysis

2.1 Is the Company Growing Month-Over-Month?

The monthly sales analysis reveals an overall upward trend across the year, though with significant fluctuations between months. The data shows a clear seasonal pattern with strong performance in Q1 (March spike), a mid-year plateau, and a significant surge in Q4 (September–November).

Monthly Growth Summary:

Month	Current Sales (\$)	Previous Sales (\$)	Growth %
February	59,751	94,925	-37.05%
March	205,005	59,751	+243.1%
April	137,762	205,005	-32.8%
May	155,029	137,762	+12.53%
September	307,650	159,044	+93.44%
November	352,461	200,323	+75.95%

Conclusion: The company shows strong growth in Q4 (Sep–Nov), confirming seasonal buying patterns. March and September are peak months. Overall, the company is growing but inconsistently — management should plan inventory and marketing around these seasonal peaks.

3. Regional Performance Analysis

3.1 Which Region is Underperforming?

Region	Total Sales (\$)	Total Profit (\$)	Status
West	725,458	108,418	Best Performing
East	678,781	91,523	Strong
Central	501,240	39,706	Underperforming
South	391,722	46,749	Needs Attention

Conclusion: The Central region is the most underperforming with only \$39,706 in profit despite \$501K in sales — indicating a very low profit margin of ~7.9%. South also underperforms relative to its sales volume. Both regions require targeted intervention.

4. Category Performance Analysis

4.1 Which Category Has the Highest Growth?

Category	Total Sales (\$)	Profit Margin %
Technology	836,154	17.4%
Office Supplies	719,047	17.04%
Furniture	742,000	2.49%

Conclusion: Technology has both the highest sales and the highest profit margin at 17.4%, making it the top-performing category. Office Supplies follows closely at 17.04%. Furniture is a major concern — despite high sales volume, its profit margin is only 2.49%, which is dangerously low.

5. Discount Impact Analysis

5.2 Are Discounts Reducing Profit?

Analysis of the dataset reveals a clear negative correlation between high discount rates and profitability. Orders classified as 'High Value' (Sales > \$1,000) show significantly lower profit margins when heavy discounts are applied. The Furniture category, which has the lowest profit margin (2.49%), also tends to carry the highest average discounts.

- Orders with 0% discount: Average profit margin ~18-20%
- Orders with 20-45% discount: Average profit margin drops below 5%
- Orders with 50%+ discount: Often result in negative profit (loss)

Conclusion: Yes — excessive discounting is directly reducing profitability, particularly in the Furniture and Office Supplies categories. A discount cap policy should be implemented.

6. Customer Segment Analysis

6.3 Which Segment Contributes Most Revenue?

Segment	Revenue Contribution	Share %
Consumer	\$1,161,401	~50.5%
Corporate	\$706,146	~30.7%
Home Office	\$429,654	~18.7%

Conclusion: The Consumer segment dominates with over 50% of total revenue, making it the most critical segment for business growth. Corporate is the second largest and represents an important B2B opportunity. Home Office, while smallest, has strong profit potential due to focused purchasing behavior.

7. Strategic Recommendations

Based on the comprehensive analysis, the following strategic actions are recommended for management:

7.1 Address Regional Underperformance

- Launch targeted sales campaigns in the Central and South regions
- Review operational costs in Central — high sales with low profit suggests cost inefficiency
- Deploy regional sales managers with specific profit margin targets

7.2 Fix Furniture Category Profitability

- Renegotiate supplier contracts to reduce cost of goods for Furniture
- Implement a minimum 15% profit margin threshold for Furniture orders
- Reduce or eliminate deep discounting on Furniture products

7.3 Discount Policy Reform

- Cap maximum discount at 20% across all categories
- Require manager approval for discounts above 15%
- Introduce loyalty rewards instead of upfront discounts to protect margins

7.4 Capitalize on Technology Growth

- Increase inventory and marketing investment in Technology products
- Expand Technology product range — highest margin at 17.4%
- Target Corporate segment with Technology bundles for B2B upselling

7.5 Seasonal Planning

- Prepare promotional campaigns ahead of September and November peaks
- Stock up on Technology and Office Supplies before Q4 rush
- Use Q2 (slower months) for operational improvements and training

8. Conclusion

The Superstore business demonstrates solid revenue generation with \$2.3M in total sales and \$286K in profit. However, profitability is unevenly distributed — Technology and Office Supplies drive strong margins while Furniture drags overall performance. The Central region requires immediate attention to improve its profitability ratio. The Consumer segment remains the backbone of the business and should be prioritized in all growth strategies.